

Guess The Number Game

Project Overview

Guess The Number is an interactive Python game where the player attempts to guess a randomly generated number between 1 and 100.

The player has 7 attempts to find the correct number. After each incorrect guess, the game provides a hint telling the player whether their guess was too high or too low.

The game also records statistics across multiple rounds, including games played, games won, games lost, win percentage and total score.

How to Play

1. Enter your name when prompted.
2. The game generates a random number between 1 and 100.
3. Enter a whole number between 1 and 100.
4. The game tells you whether your guess is too high or too low.
5. Continue guessing until you find the number or use all 7 attempts.
6. If you win, you receive points based on the number of attempts used.
7. Choose whether you would like to play another round.
8. When you choose No, your final statistics are displayed.

Game Features

- Random number generation
- 7 attempts per round
- High and low hints
- Input validation
- Error handling for invalid and non-numeric input
- Scoring system
- Multiple rounds
- Games won and lost tracking
- Win percentage calculation
- Total score tracking

Input Validation and Error Handling

The game checks that the player's input is a valid whole number.

A try and except statement is used to prevent the program from crashing when the player enters invalid input, such as letters or other non-numeric values.

The game also checks that the number entered is between 1 and 100. If it is outside this range, the player is asked to enter another number.

Functions

The program is divided into three main functions.

`get_guess()`

Gets the player's guess and continues asking until a valid number between 1 and 100 is entered.

`game()`

Controls one round of the game, including generating the secret number, managing attempts, checking guesses and updating the game statistics.

`statistics()`

Displays the player's final statistics, including games played, games won, games lost, win percentage and total score.

Scoring

The player receives more points for guessing the number using fewer attempts.

The maximum score is awarded for correctly guessing the number on the first attempt, while fewer points are awarded for taking more attempts.

Game Statistics

The program records:

- Games Played
- Games Won
- Games Lost
- Win Percentage
- Total Score

These statistics are displayed when the player chooses to stop playing.

Technologies Used

Programming Language: Python

Module Used: random

The random module is used to generate the secret number between 1 and 100.

How to Run

The Python file can be run using a Python development environment such as VS Code.